

Here is the condensed, high-priority safety and tools checklist for PC assembly and disassembly:

Essential Safety Precautions

- **Power Down:** Turn off the PC, flip the PSU switch to "O" (Off), and completely unplug the main power cord. This eliminates the risk of fatal short circuits or electrical shocks.
- **Drain Residual Energy:** Hold down the case's power button for 5–10 seconds after unplugging to safely empty the motherboard's capacitors.
- **Avoid Static (ESD):** Work on a hard, non-conductive surface (like wood) rather than a carpet. Frequently touch the bare metal of the PC chassis, or wear an anti-static wrist strap, to avoid frying delicate chips with static electricity.
- **Handle with Care:** Always hold components (RAM, GPU, Motherboard) by their plastic or metal edges. Never touch the exposed gold pins or printed circuit board (PCB) faces directly to prevent damage from skin oils.

Required Tools

- **Phillips-Head #2 Screwdriver:** The universal tool for almost every PC screw. A magnetized tip is highly recommended to catch tiny screws before they fall into tight crevices.
- **Sorting Tray/Cups:** Essential for keeping different screw types separate (e.g., coarser PSU/case screws vs. finer motherboard screws) so you don't strip the threading.
- **Anti-Static Bags:** Safe storage to protect loose, highly sensitive components from environmental static and dust while they are outside the case.
- **Zip Ties / Velcro Straps:** Used to secure loose cables out of the way, keeping them clear of spinning fan blades and ensuring optimal system airflow.